

Project Build Summary

Sufi Mahbub Ahmed — Cybersecurity Portfolio Website

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Institution: Islamic University of Technology (IUT)

Host: GitHub Pages

1. EXECUTIVE OVERVIEW

This document summarizes the end-to-end architecture, development workflow, design decisions, and deployment steps undertaken to build the personal cybersecurity portfolio website for **Sufi Mahbub Ahmed** from scratch.

Live Deployment URL: <https://nirjhorahmed.github.io/Portfolio/>

GitHub Repository: <https://github.com/NirjhorAhmed/Portfolio>

2. STEP-BY-STEP DEVELOPMENT LIFECYCLE

1 Requirements Analysis & Compliance Formulation

Analyzed `portfolio.txt.txt` to extract all structural constraints. Enforced honest student representation: strictly omitting fabricated work experience, fake internships, false certifications, graduation dates, or LinkedIn links, while transforming services into "Areas of Future Focus".

2 Asset Management & Visual Identity Design

Created the directory hierarchy (`assets/images/`, `css/`, `js/`, `.github/workflows/`). Integrated the user-uploaded portrait (`profile.jpg`) and styled it with a high-tech glowing orbit frame, cyber brackets, and floating status chips.

3 Semantic HTML5 Front-End Construction

Developed `index.html` with all core sections: Sticky Navigation, Hero section, About Me, Education Timeline (IUT, Dhaka Commerce College HSC 5.00, MUBC SSC 5.00), Categorized Technical Skills, Projects & Learning Roadmap, Career Journey, Areas of Future Focus, Gaming Interests, and Contact form with Email Copy button.

4 Cyber-Dark Styling & Animations Engine

Crafted `css/style.css` and `css/animations.css` using modern CSS variables, glassmorphism (`backdrop-filter`), cyber-neon cyan highlights (`#00f0ff`), responsive flex/grid layouts, glowing borders, and scroll-reveal triggers.

5 Interactive 3D Cyberspace "Entering The System" Engine

Engineered a custom 60 FPS HTML5 Canvas engine in `js/particles.js`. Implemented a 3D perspective grid highway (floor & ceiling), passing security firewall gates (`SYS_AUTH`, `FIREWALL // BYPASS OK`, `MAINFRAME // LEVEL 01`), peripheral matrix code drops, interactive mouse flight steering, and scroll speed acceleration.

6 Git Version Control & GitHub Pages CI/CD Deployment

Configured git tracking, added `.nojekyll` for raw static file serving, created automated GitHub Actions deployment workflow (`.github/workflows/static.yml`), and pushed all assets to the public GitHub repository for live hosting.

3. TECHNICAL ARCHITECTURE & FILE STRUCTURE

File / Path	Type	Primary Role & Functionality
index.html	HTML5	Single-page semantic structure, accessibility attributes, meta tags, and section anchors.
css/style.css	CSS3	Dark cyber theme, custom properties, responsive breakpoints, card glassmorphism.
css/animations.css	CSS3	Keyframe glow pulses, floating chip animations, orbit rotations, scroll reveal styles.
js/particles.js	Vanilla JS	3D Cyberspace matrix highway, security gate portals, perspective projection math.
js/main.js	Vanilla JS	Scrollspy active tracking, sticky nav, mobile menu drawer, email clipboard copy, toast.
assets/images/profile.jpg	JPEG Image	User portrait featured in hero cyber frame with glow borders and corner brackets.
.github/workflows/static.yml	YAML CI/CD	Automated deployment pipeline building and deploying website to GitHub Pages.
README.md	Markdown	Repository documentation, features summary, and step-by-step setup guides.

4. REQUIREMENTS COMPLIANCE VERIFICATION

Theme & Identity Passed Deep cyber-blue palette, neon cyan highlights, clean minimal layout, subtle animations, student cybersecurity identity.	Honest Representation Passed No fabricated jobs, companies, internships, fake graduation dates, or services. Portrayed strictly as an active IUT student.
Contact & Social Rules Passed Includes email, phone, GitHub profile, and interactive form. LinkedIn link strictly omitted as requested.	Extensibility & Hosting Passed Fully modular architecture, GitHub Pages static hosting ready, and extensible for future semester projects.